

on adding features and designing effective user interfaces for devices such as thermostats, industrial panels, and other interactive displays.

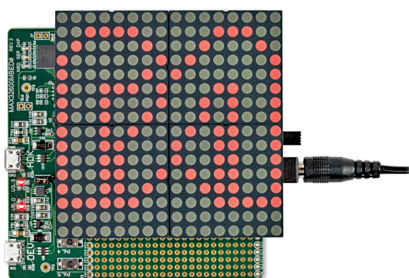
Applications. Suitable for devices requiring a touchscreen or graphical interface

Link. <https://www.electronicsforu.com/electronics-projects/gui-reference-design-for-embedded-display-applications>

OEM Brand. Nuvoton

Display driver shield

Small 8×8 LED matrices often make characters difficult to read, and connecting multiple displays can require excessive wiring and introduce data reliability issues.



Voltage compatibility across different microcontrollers and limited power capacity can also reduce stability, slowing testing and introducing errors. This reference design combines four 8×8 displays into a 16×16 matrix, making characters bigger and wiring easier with chained drivers. It supports standard digital interfaces, works with both 3.3V and 5V systems, uses an external power supply for improved reliability, and includes ready-to-use code for popular platforms. This allows us to focus on building devices rather than developing low-level drivers. The compact, cost-effective design simplifies prototyping and reduces development effort.

Applications. Suitable for signs, interactive displays, status panels, shop messages, and other projects requiring clear, flexible LED displays

Link. <https://www.electronicsforu.com/electronics-projects/display-driver-shield-reference-design>

OEM Brand. Analog Devices (ADI)

Embedded touchscreen GUI

This GUI reference design enables us to add touchscreens and graphics to embedded devices without additional cost or complexity. It supports responsive touchscreen interfaces using a single 32-bit microcontroller for both display and touch,



removing the need for separate touch hardware. The design provides ready-made hardware, software, and adaptable code layouts that help reduce development time and risk while improving reliability. It can be used for consumer devices such as smart appliances and personal electronics, as well as industrial systems including control panels, HMIs, automotive dashboards, and medical devices. Integrated tools support interface design, code generation, consistency, and scalability for graphics-enabled embedded products.

Applications. Suitable for industrial, consumer, and smart-connected applications

Link. <https://www.electronicsforu.com/electronics-projects/accelerating-embedded-touchscreen-gui-reference-design>

OEM Brand. Microchip

Automotive infotainment display module

Automotive display modules

require power systems that operate reliably under demanding conditions, including weak battery starts, sudden voltage drops, and voltage spikes. They must also comply with safety standards while supplying multiple



voltage rails for backlights, display controllers, and communication interfaces, all within a compact, efficient design that minimises interference. This reference design addresses these requirements in three ways. It protects the system from unsafe input voltages and overcurrent, boosts a weak battery voltage to ensure reliable start-up, and delivers stable power to all display components. It also monitors voltage rails for faults and enables safe recovery, supporting compliance with safety standards.

Applications. Suitable for reliable, safety-compliant automotive display modules

Link. <https://www.electronicsforu.com/electronics-projects/automotive-infotainment-display-module-reference-design>

OEM Brand. Texas Instruments (TI)

These reference designs allow us to combine proven solutions, solve integration challenges quickly, and develop display systems more efficiently. They provide a practical path to faster and more reliable display implementation. **EFY**

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